TELECOMMUNICATIONS PRIMER · MOBILE NETWORK FUNDAMENTALS

Public Land Mobile Network (PLMN)

A Public Land Mobile Network (PLMN) is a wireless communication network operated by a mobile network operator to provide voice, messaging, and data services within a geographical area.



01 Main components

COMPONENT 01
Mobile Station (MS)

The mobile phone or user device. Sends and receives voice and data.

COMPONENT 02
Base Station Subsystem

BTS and BSC in traditional GSM. Provides radio communication between devices and the network.

COMPONENT 03
Network Switching Subsystem

MSC, HLR, VLR and AuC. Manages call routing, subscriber information and authentication.

COMPONENT 04
External Networks

Links to other mobile networks and the PSTN, enabling calls outside the home network.

02 Simplified architecture

TRADITIONAL GSM MODEL



03

MCC — COUNTRY
Mobile Country Code
Identifies the country.

MNC — OPERATOR
Mobile Network Code
Identifies the operator.

PLMN Identity = MCC + MNC

04 How a PLMN works

From network discovery to an active voice or data session

1. **Search** — Device scans for available networks.
2. **Select** — A suitable PLMN is selected.
3. **Authenticate** — Subscriber identity is verified.
4. **Connect** — Voice or data link is established.
5. **Communicate** — Traffic flows via the mobile network.

NETWORK SELECTION — SERVICE DELIVERY

05 Benefits & applications

- **Voice calls and SMS** — Reliable person-to-person services
- **Mobile internet access** — Always-on data connectivity
- **Subscriber authentication** — Secure access to operator services
- **Mobility and roaming** — Service continuity while moving
- **Wide-area connectivity** — Coverage across geographical areas